



Firefox experiments with Blekko

In an effort to integrate a search engine like its rivals, Mozilla has entered into partnership with instant search engine, Blekko <http://dgit.in/Lq5MAg>

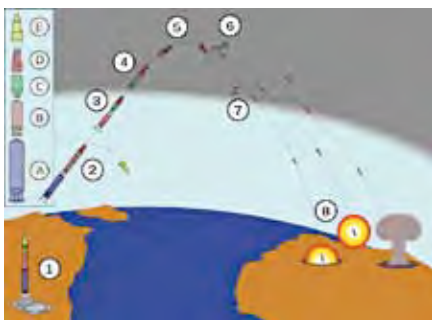
Socialize while dying

My Last Wish is a social network which helps people to socialize with other dying people that have similar last wishes <http://dgit.in/KeWJof>

like the obvious choice then. As Avinash Chander, director of ASL describes, “The Agni-V is specially tailored for road-mobility. With the canister having been successfully developed, all India’s future land-based strategic missiles will be canistered as well.”

The most important aspect of the Agni V is the incorporation of multiple independent re-entry vehicle or MIRV technology. It’s best to think of this as a guided cluster bomb – the main rocket moves into suborbital atmosphere and pushes a bus containing the nuclear warheads into a free-flight path. From here, it becomes a re-entry vehicle that manoeuvres using on-board systems to different targets, releasing a warhead on the calculated trajectory. The process is repeated for multiple targets. What makes MIRV so dangerous is that (a) anti-ballistic missile systems become almost useless, since they’re designed to attack a singular warhead with a singular missile (b) can cause a large amount of damage to a number of targets, while only requiring a single missile for launch, and (c) increase the effectiveness of radiation, since it counters the inverse-square law (where radiation diminishes as distance from target increases).

In addition to inertial navigation systems, the Agni V is also equipped with a ring laser gyroscope for accurate guidance



Stages of an MIRV's working, from launch to suborbital height and re-entry, ending in deployment of warheads

towards a target. This is made possible by using a ring laser, which detects shifts in standing wave patterns between the counter-operating beams, thus observing rotation. Ring laser gyroscopes has no moving parts, and are therefore more accu-

rate. It doesn't suffer orientation changes, and is tough, deployed in most commercial, military aircraft, and spaceships.

“MIRV technology is similar to Indian Space Research Organization’s multiple satellite injection technology. The defence scientists have to militarise it and install better guidance system to ensure that warheads hit the targets,” according to Agni Project Director Tessy Thomas.

Each Agni V missile will be capable of carrying between 2 to 10 separate nuclear warheads; using different on-board inertial guidance systems and rocket motors; each warhead can be steered to different targets, hundreds of kilometres away from each other, thus maximizing the damage caused by a single missile. Conversely, a select few warheads can assigned to a single target. Deeming it a “force multiplier,” Saraswat explains that, “Where I was using four missiles, I may only use one missile. So it becomes a force multiplier given the damage potential.” This single aspect alone has catapulted India into a league occupied by the likes of the United States, Russia, China, France and Britain. MIRV is still currently under development at the DRDO, and the exact date of its testing is still unknown.

India's Future in Missile Technology

With so much destructive power already at its capability what comes next? The obvious indication is that the DRDO would begin developing missiles with ranges of 8,000 kilometres and more. After all, China’s ICBM capability is significantly larger. “Whenever the need arises, we will develop. Right now we feel, 5,000 km-class of system meets our threat perception and requirement. But that doesn't mean we won't develop a longer range missile,” Saraswat said.

This development seems to be heading towards furthering the intercontinental divide, most specifically with the Agni VI. As of now, there is no official confirmation from the DRDO on the existence of the Agni VI, though a newsletter published last year alluded to the existence of an ICBM with a range of 6,000-10,000 kilometres. This missile would also have the capability of launching from underwater,

as well as from the ground, and would be equipped with MIRV technology.

Other missile technologies the DRDO are developing include the K-X series of missiles, designed to launch from submarines. It features three variants: The K-15 (surface-based) which has a range of 750 kilometres, the K-4, which is based off the Agni III/IV technology



Comparison of missile ranges for Agni II, Agni III and Agni V

with a range 3,500-4,500 kilometres and the upcoming K-15, based off the Agni V. This is not including the Astra family of missiles, a beyond-visual range, air-to-air missile which is to be fitted on different fighter aircraft. The Astra Mark 1 missile was tested last year by the DRDO, and can reportedly annihilate targets at supersonic speeds and at ranges of 21-110 kilometres; depending on the altitude it is fired. It reportedly has an active homing range of 25 kilometres, and contains on-board electronic counter counter-measures or ECCMs to jam radar signals from Surface-to-Air Missile or SAM sites. This means it can't be shot down easily either. The Mark 2 version of the Astra reportedly has a range of 150 kilometres, though Saraswat would not comment on its actual range. On a side note, there's also ISRO's 640 ton GSLV MK3 rocket (dubbed the “Monster Rocket”), India's bid for launching astronauts into space.

However one may feel about nuclear proliferation, Agni V is a proud moment for us. It's a little disheartening that its electronics were imported from outside, indicating we still lag behind in manufacturing computers that can match developed nations' military tech. Nonetheless, the Agni-V represents a significant step towards increasing security for India. **4**